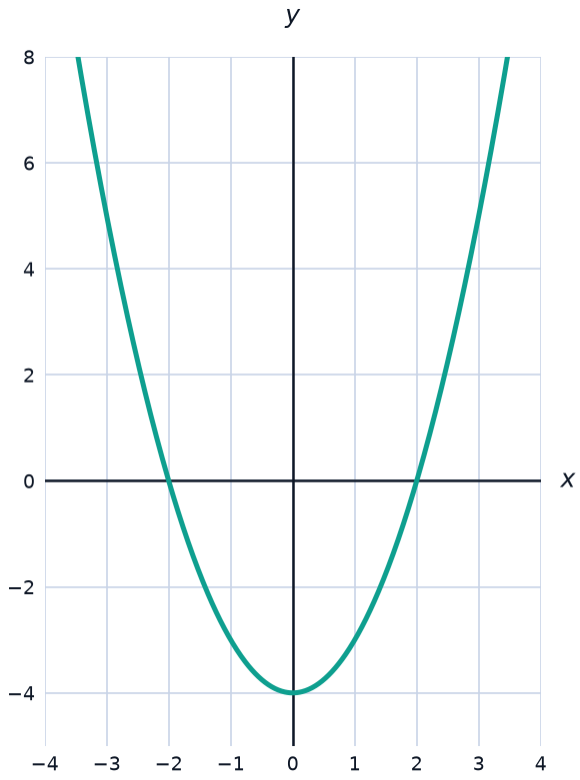
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Reading a function graph**

The graph of $y = f(x) = x^2 - 4$ is shown.



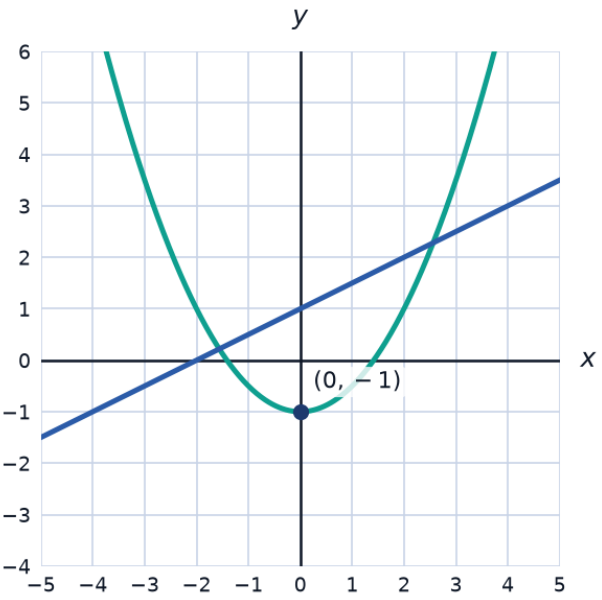
a State the zeros of f.

b State the domain and range of f.

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2 [Foundational] • Target: Evaluating functions

If $f(x) = 3x - 4$, find $f(5)$.



3 [Foundational] • Target: Evaluating functions

If $f(x) = x^2 - 2x + 1$, find $f(-2)$.

4 [Foundational] • Target: Evaluating functions

If $g(x) = \sqrt{x + 5}$, find $g(4)$.

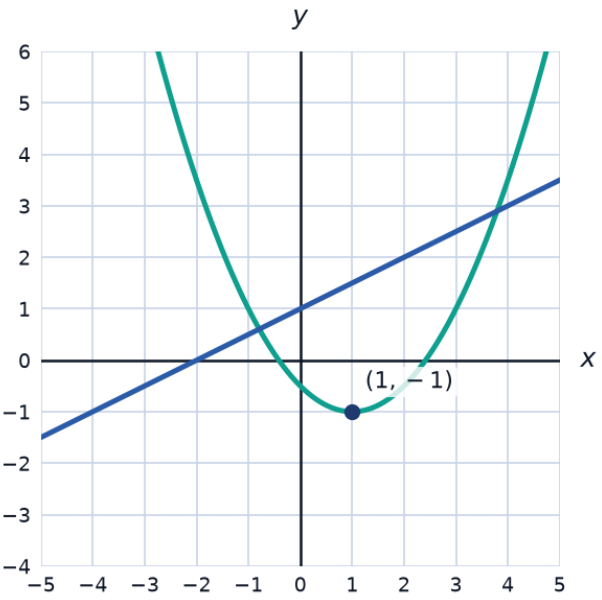
5 [Foundational] • Target: Evaluating functions

If $h(x) = \frac{2x}{x - 1}$, find $h(3)$.

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6 [Foundational] • Target: Domain and range

State the domain of $f(x) = \sqrt{x - 3}$.



7 [Foundational] • Target: Domain and range

State the domain of $f(x) = \frac{1}{x - 5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range

State the domain of $f(x) = \sqrt{7 - x}$.

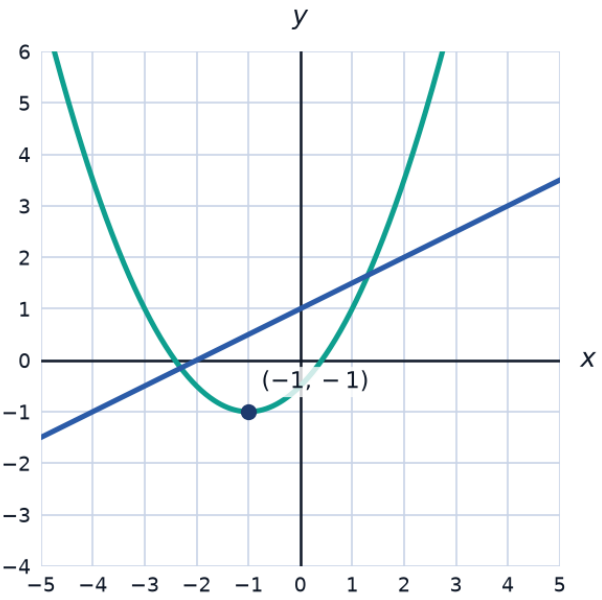
9 [Intermediate] • Target: Domain and range

State the range of $f(x) = x^2 + 2$.

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10 [Intermediate] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.



11 [Intermediate] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.

12 [Intermediate] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

13 [Intermediate] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.

14 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Transformations: reflections and stretches • Cross-Lesson: 3.01 Sequences & Series

Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

16 [Challenging] • Target: Composite functions • Cross-Lesson: 4.01 Trigonometry

Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.

17 [Challenging] • Target: Composite functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Using the same f and g, find $(g \circ f)(2)$.

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18 [Challenging] • Target: Composite functions • Cross-Lesson: 6.01 Limits & Derivatives

Using the same f and g, find a general formula for $(f \circ g)(x)$.

19 [Challenging] • Target: Composite functions • Cross-Lesson: 7.01 Integration & Applications

Using the same f and g, find a general formula for $(g \circ f)(x)$.

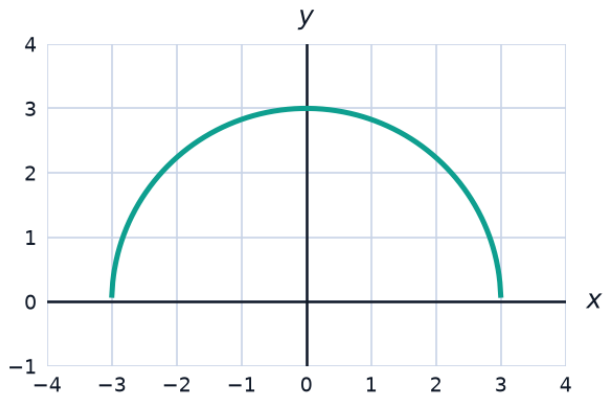
20 [Challenging] • Target: Inverse functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Find the inverse of $f(x) = 3x - 2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Domain and range

The graph shows $f(x)$ defined only where plotted. State its domain.



a. $[-3, 3]$

b. All real numbers

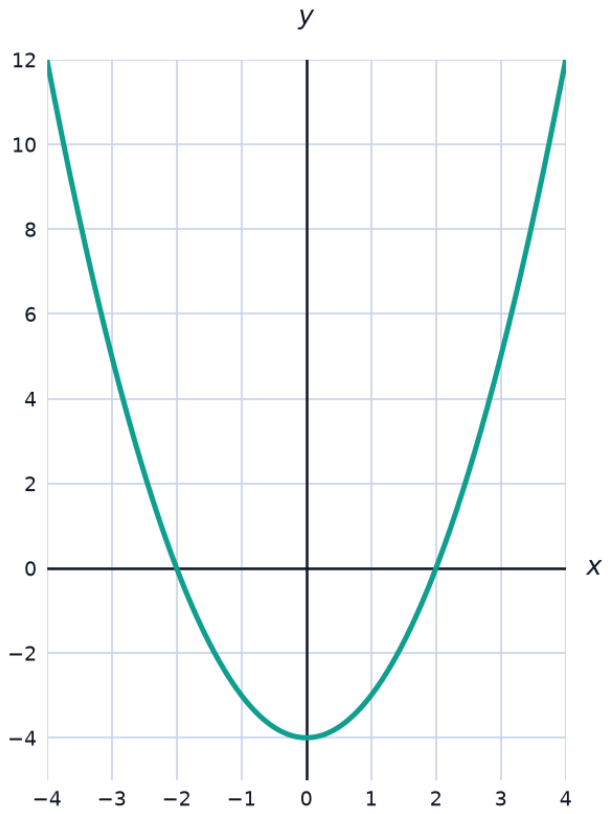
c. $[-4, 4]$

d. $[0, 3]$

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22 [Foundational] • Target: MCQ — Even and odd functions

The graph shown is symmetric about the y-axis. Is the function even, odd, or neither?



a. Odd

b. Neither

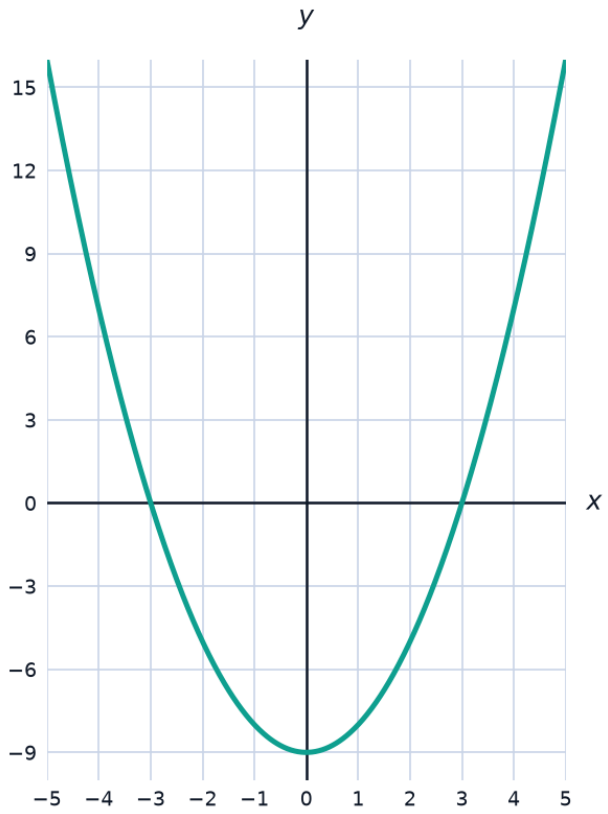
c. Cannot be determined

d. Even

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23 [Foundational] • Target: MCQ — Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.



a. -9

b. 6

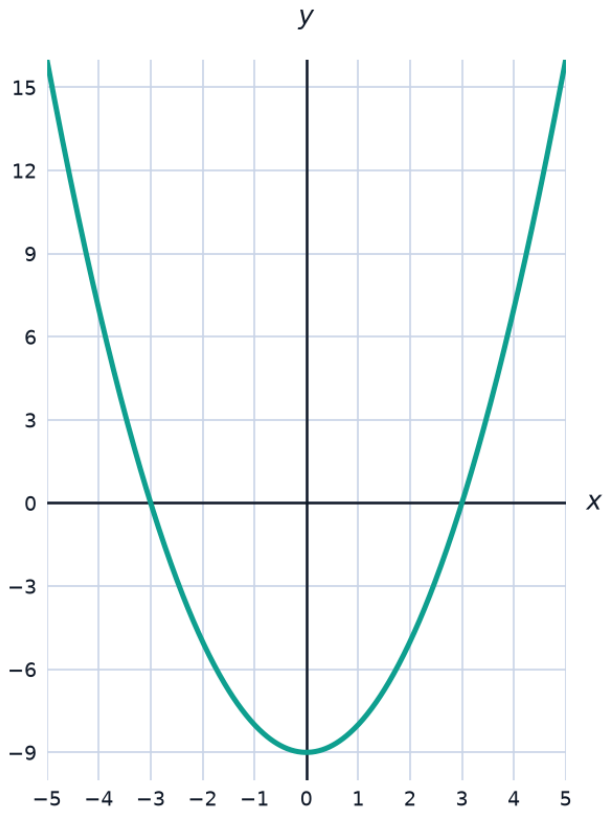
c. 9

d. 0

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24 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the values of x where $f(x) = 0$ (the x-intercepts).



a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

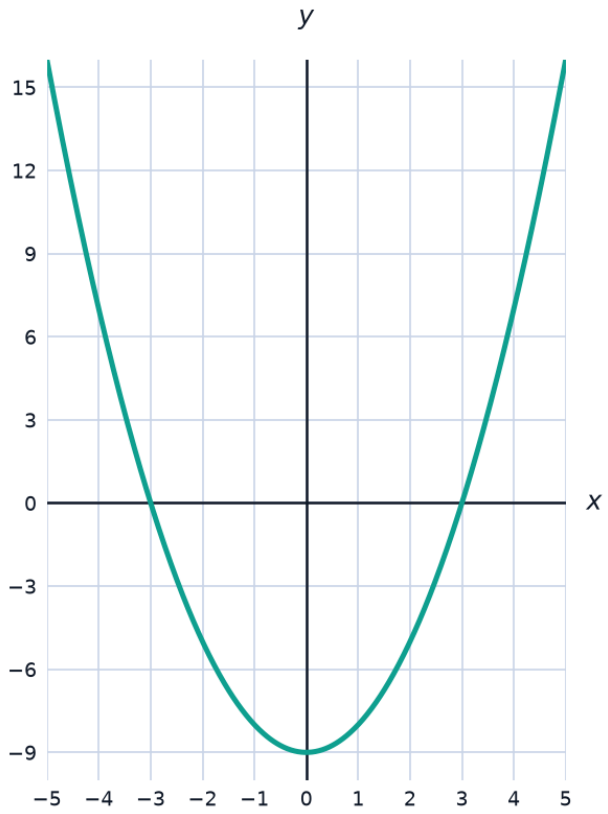
c. $x = 3$ only

d. $x = -3$ and $x = 3$

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25 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the y-intercept.



a. -3

b. 9

c. 0

d. -9

26 [Foundational] • Target: MCQ — Evaluating functions

If $f(x) = 4x - 7$, find $f(6)$.

a. 17

b. 24

c. 31

d. 10

27 [Foundational] • Target: MCQ — Evaluating functions

If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16

d. 22

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28 [Foundational] • Target: MCQ — Domain and range

State the range of $f(x) = x^2 + 5$.

a. $y > 5$

b. All real numbers

c. $y \leq 5$

d. $y \geq 5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 2. Write the new equation.

a. $y = -\frac{1}{2}x^2$

b. $y = (-2x)^2$

c. $y = -2x^2$

d. $y = 2x^2$

30 [Intermediate] • Target: MCQ — Composite functions

Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $(f \circ g)(3)$.

a. 26

b. 23

c. 11

d. 29

31 [Intermediate] • Target: MCQ — Inverse functions

Show that $f(x) = 3x + 9$ and $g(x) = \frac{x-9}{3}$ are inverses by computing $f(g(x))$.

a. x

b. $x + 9$

c. $3x$

d. $x - 9$

32 [Intermediate] • Target: MCQ — Piecewise functions

Using the same piecewise function, find $f(2)$.

a. 3

b. 4

c. 0

d. 5

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33 [Intermediate] • Target: MCQ — Even and odd functions

Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

a. Both even and odd

b. Neither

c. Odd

d. Even

34 [Intermediate] • Target: MCQ — Increasing, decreasing, and average rate of change

Find the average rate of change of $f(x) = x^3$ from $x = 1$ to $x = 3$.

a. 8

b. 26

c. 9

d. 13

35 [Intermediate] • Target: MCQ — Word problems: functions in context

A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 dollars

b. 150 dollars

c. 600 dollars

d. 900 dollars

36 [Intermediate] • Target: MCQ — Transformations: reflections and stretches

If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3

d. 3

37 [Intermediate] • Target: MCQ — Composite functions

If $f(x) = x^2 + 8x - 1$, find $f(-5)$.

a. -19

b. -16

c. -13

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

If $f(x) = x^2 + 6x + 9$, find $f(-5)$.

a. 4

b. 1

c. 7

d. -4

39 [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

If $f(x) = x^2 - 2x - 2$, find $f(-2)$.

a. 9

b. -6

c. 3

d. 6

40 [Challenging] • Target: MCQ — Even and odd functions • Cross-Lesson: 8.01 Exponentials & Logarithms

State the domain of $f(x) = \sqrt{3x + 6}$.

a. $x \geq -2$

b. $x \geq -4$

c. $x \leq -2$

d. $x \geq 0$

41 [Challenging] • Target: MCQ — Increasing, decreasing, and average rate of change • Cross-Lesson: 9.01 Statistics & Probability

State the domain of $f(x) = \sqrt{x - 2}$.

a. $x \leq 2$

b. $x \geq 4$

c. $x \geq 0$

d. $x \geq 2$

42 [Challenging] • Target: MCQ — Operations on functions • Cross-Lesson: 1.01 Algebra & Equations

State the domain of $f(x) = \sqrt{-2x + 14}$.

a. $x \leq 7$

b. $x \leq 5$

c. $x \geq 7$

d. $x \leq 9$

43 [Challenging] • Target: MCQ — Reading a function graph • Cross-Lesson: 3.01 Sequences & Series

Let $f(x) = -x - 2$ and $g(x) = -x^2 + 3$. Find $(f \circ g)(1)$.

a. -4

b. -7

c. -1

d. 4

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44 [Challenging] • Target: MCQ — Word problems: functions in context • Cross-Lesson: 4.01 Trigonometry

Find the inverse of $f(x) = 5x + 1$.

a. $\frac{x-1}{5}$

b. $\frac{x-1}{-5}$

c. $\frac{x+1}{5}$

d. $5x + 1$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = |x + 7| + 3$

b. $y = |x - 7| - 3$

c. $y = |x + 7| - 3$

d. $y = |x - 7| + 3$

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